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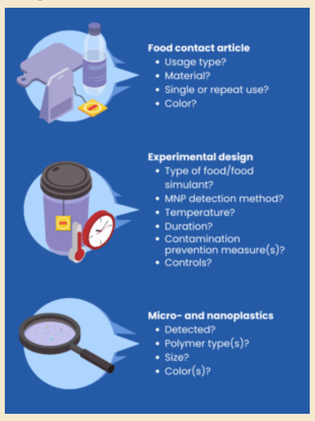


# *Release mechanisms, analytical evidence and risk implications*


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**Overview:** The article synthesises EFSA 2025 evidence on the release of micro- and nanoplastics from food contact materials. Common packaging polymers such as PE, PP, PET, PS and PLA can emit particles during production, handling, heating and storage. Reported release levels typically range from $10^2$ to $10^6$ particles per litre, while EFSA highlights the urgent need for harmonised methods, certified reference materials and integrated risk assessment.



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## BACKGROUND AND SCOPE

Micro- and nanoplastics are increasingly recognised as contaminants in food systems, and food contact materials represent a direct route of dietary exposure. The review screened studies from 2018-2024 that investigated polymer degradation, analytically confirmed polymer identity and reported quantitative or semi-quantitative release data. Approximately 130 studies met the inclusion criteria.

The article also places the EFSA 2025 review in a broader European context by linking analytical characterisation, exposure quantification, toxicological evidence and risk-management needs into one harmonised framework for food safety assessment.

## MAIN RELEASE MECHANISMS

MNP detachment is driven mainly by thermo-oxidation, hydrolysis, mechanical abrasion and photolytic degradation.

• PE and PP: repetitive heating, UV exposure and mechanical stress promote chain scission and cracking.
• PET: hydrolytic cleavage of ester bonds is favoured by moist, acidic and elevated-temperature conditions.
• PS: photo-oxidation produces brittle oxidised layers that fragment under shear stress.
• PLA: despite being biodegradable, it can still emit fragments during thermal or enzymatic degradation.

Across studies, the highest emissions were associated with longer contact time, repeated reuse and higher temperatures.

## ANALYTICAL EVIDENCE, QUALITY ASSURANCE AND METHOD PERFORMANCE

Robust evidence for MNP release comes from vibrational spectroscopy, imaging tools and thermo-analytical methods. Micro-FTIR and Raman provide polymer-specific identification; LDIR accelerates mapping and automatic particle classification; while Py-GC/MS and TED-GC/MS support mass-based quantification in complex mixtures.
Reliable studies use HEPA-filtered laminar flow conditions, non-plastic laboratory equipment, procedural blanks and recovery experiments. However, incomplete QA/QC reporting remains a major weakness in the literature. The review emphasises that certified reference materials, shared spectral databases and inter-laboratory validation are essential for comparable and traceable results.

## MATRIX EFFECTS AND INDICATIVE RELEASE LEVELS

Most investigations report release within $10^3$-$10^5$ particles $L^{-1}$, while the wider evidence base spans roughly $10^2$-$10^6$ particles $L^{-1}$ depending on polymer type and conditions. Examples reported in the review include:
PET - water or soft drink, 90 °C / 24 h: about 5 x $10^4$ particles $L^{-1}$
PP - milk or water, 70 °C / 1 h: about 3 x $10^4$ particles $L^{-1}$
PE - acidic simulant, 60 °C / 6 h: about 2 x $10^4$ particles $L^{-1}$
PS - boiling water, 100 °C / 30 min: about 1 x $10^4$ particles $L^{-1}$
PLA - hot tea, 80 °C / 10 min: about 0.8 x $10^4$ particles $L^{-1}$

## RISK ASSESSMENT FRAMEWORK

EFSA's holistic framework links analytical detection and exposure estimation with hazard identification, characterisation and risk management. Current toxicological evidence suggests possible oxidative stress, inflammation and membrane disruption, but human-relevant dose-response relationships remain insufficiently defined. Nanoplastics are of particular concern because particles below 1 µm may cross epithelial barriers or interact with immune cells.

## PRIORITIES FOR FUTURE ACTION

The article identifies several priorities for the next phase of research and regulation:

• harmonised digestion, filtration and reporting protocols
• polymer-specific certified reference materials for PE, PP, PET, PS and PLA
• inter-laboratory validation coordinated within European reference frameworks
• greater use of mass-based metrics in addition to particle counts
• advanced methods such as AFM-IR, LDIR and SP-ICP-MS for submicrometre particles
• innovation in low-migration and sustainable packaging materials

## CONCLUSION

Food contact materials are a measurable source of micro- and nanoplastic contamination in food systems. Release intensity depends on the polymer type, contact medium, surface damage, time and temperature. Despite major analytical advances, the lack of harmonised international protocols and certified reference materials still limits comparability and slows risk assessment.
The review therefore supports a combined strategy of better analytics, coordinated monitoring, realistic exposure modelling and stronger material innovation to reduce contamination at source and improve consumer protection.

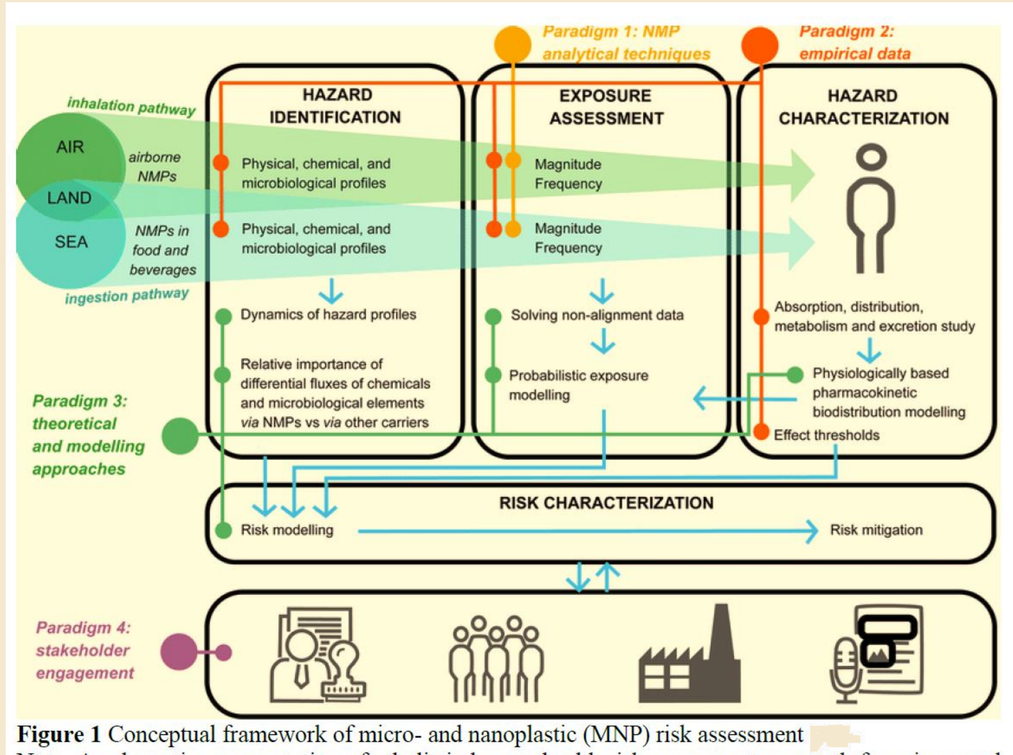


**Figure 1** Conceptual framework of micro- and nanoplastic (MNP) risk assessment
Note: A schematic representation of a holistic human health risk assessment approach for micro- and nanoplastics, integrating four complementary paradigms:
(1) analytical techniques and exposure quantification,
(2) empirical toxicological evidence,
(3) theoretical and probabilistic modelling approaches, and
(4) stakeholder engagement.

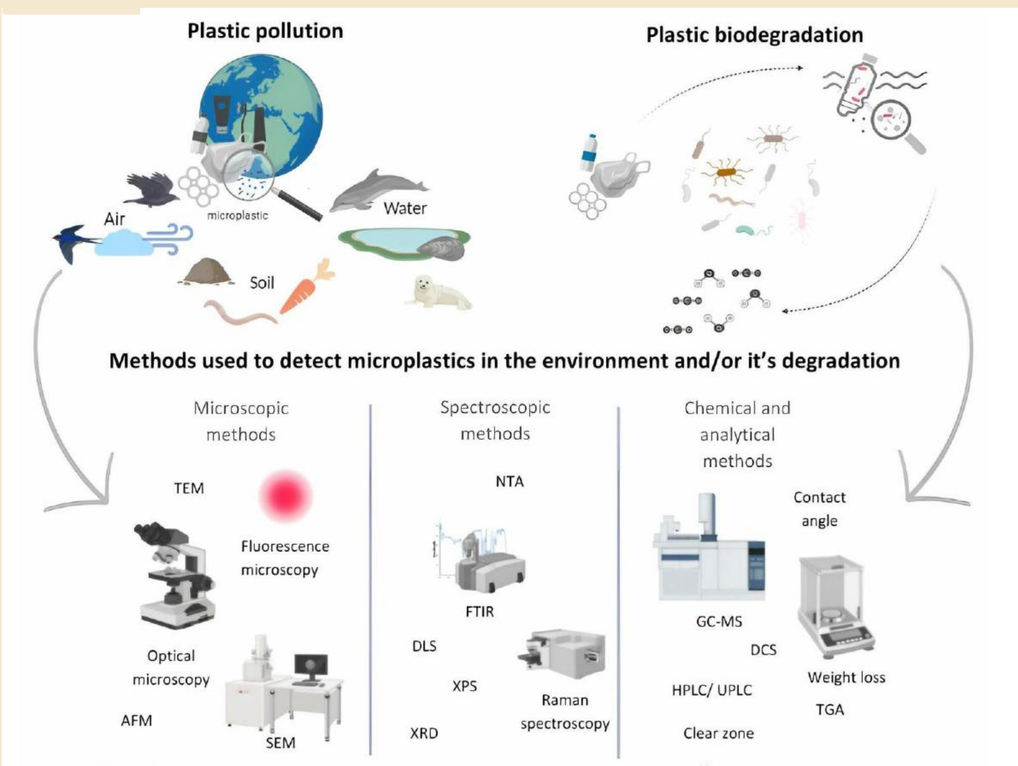


**Figure 2** Analytical workflow for micro- and nanoplastic identification


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